ECO 362 Financial Investments
Final Exam

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Instructions

1. Check that you have all 4 pages. There are 6 questions.

Question	Points
1	12
2	16
3	20
4	16
5	20
6	16
Total	100

2. This is a closed book exam. However, you may use a two-page cheat sheet containing formulas or anything else you may find helpful. The cheat sheet must be letter size (8.5x11in) with writing on both sides. You are under the Honor Code to comply with this requirement.

3. You may use a writing instrument, a cheat sheet, and a calculator. Put everything else away. The calculator cannot be a phone, a tablet, or a laptop. If you do not have a calculator, you may borrow one. Please return it after the exam.

4. You may not communicate with other students or violate the Honor Code during this exam. Write your name and sign the Honor Code on the first page of the exam book:

   I pledge my honor that I have not violated the Honor Code during this examination.

5. Write your answers in the exam books. *Show all steps necessary in deriving your solution for any credit.* Report the yield, the coupon rate, or the return in percentage points. Round all answers to two decimal places (e.g., $95.67 or 1.23%).

6. You have 3 hours to complete the exam. After 3 hours, you must stop working on the exam. You then have an extra 20 minutes to submit your answers as one scanned pdf file or photographed images through Gradescope. If you do not have a device to scan or photograph, give your exam books to a preceptor, who will be able to do it for you.

7. In addition to submission by Gradescope, submit the hard copies of the exam books before you leave the room. We need them in case there are any technical issues (e.g., unreadable scans). Make sure that your name is on the exam books that you submit.

1. Give short answers to each part.

   [12 points: 6 points each part]

   (a) Give two reasons why you may want to opt out of a target date fund (e.g., Fidelity Freedom Funds) as a default option in a 401(k) account and instead manage your own portfolio.

   (b) Give two reasons why you may want to contribute to a Roth 401(k) account instead of a traditional 401(k) account.

2. The following table reports the annual endowment returns for the fiscal year that ended in June 2024. The benchmark return for university endowments was 10.1%. The annually compounded riskless interest rate was 5%.

University	Return (%)
Columbia	11.5
Brown	11.3
Harvard	9.6
Cornell	8.7
Dartmouth	8.4
Penn	7.1
Yale	5.7
Princeton	3.9

   Assume that the return for university i in year t satisfies

$$R_{i,t} - R_{b,t} = \alpha_i + \beta_i(R_{m,t} - R_{b,t}) + \epsilon_{i,t},$$

   where $R_{m,t}$ is the benchmark return and $R_{b,t}$ is the riskless interest rate. Assume that $\text{Var}(\epsilon_{i,t})$ is constant across university endowments.

   [16 points: 4 points for each part]

   (a) Give two possible reasons why Princeton's endowment had a lower return than its peers. To receive credit, you must use the words "systematic risk" and "idiosyncratic risk" and relate these concepts to the variables or parameters in the equation above.

   (b) Assume that Princeton's endowment has a positive beta with respect to the benchmark return. Can lower systematic risk be the only reason for Princeton's relatively low performance?

   (c) Based only on the information given, can you definitively conclude that Columbia's endowment is better managed than Princeton's? Explain the reasoning for your answer.

   (d) Based only on the information given, estimate the idiosyncratic standard deviation of endowment returns. For this calculation, you may assume that all Ivy endowments have the same beta with respect to the benchmark return.

3. The following table reports the year-end stock price and dividend per share for Nvidia. The stock price is the ex-dividend price, meaning that \$58.82 is the resale price after you receive the dividend of \$0.16 on 12/31/2019.

Date	Stock price (\$)	Dividend (\$)
12/31/2020	130.55	0.16
12/31/2021	294.11	0.16
12/31/2022	146.14	0.16
12/31/2023	495.22	0.16

The annually compounded riskless interest rate is 5%, and the expected annual return on the overall stock market is 9%.

[20 points: 4 points for each part)]

(a) What is the annually compounded return on Nvidia from 12/31/2020 to 12/31/2021?

(b) What is the cumulative return on Nvidia from 12/31/2020 to 12/31/2023? Assume that all dividends are reinvested in Nvidia.

(c) Assume that the CAPM holds. Nvidia has a market beta of 2. What is its expected annual return?

(d) Use the Gordon growth model to calculate the expected annual dividend growth on 12/31/2023.

(e) Continue to assume the same expected annual stock return from part (c). Suppose that investors lower their expected annual dividend growth by 0.1 percentage points relative to your answer in part (d). What would be the stock price on 12/31/2023 under this lower expected dividend growth? Based on your answer, would you say that Nvidia's stock price has high or low sensitivity to the expected growth of the AI industry? Explain the reasoning for your answer.

4. You have \$1 million in financial wealth. Your current annual income is \$50,000, and you expect income to grow 2% annually. Assume a risk aversion parameter of $\gamma = 10$.

The annually compounded riskless interest rate is 5%. Annual stock returns have a mean of 9% and a standard deviation of 20%.

[16 points: 4 points for each part]

(a) Use the Gordon growth model to approximate your human capital. It is an approximation because you ignore retirement (i.e., zero income in the future) for the purposes of this calculation.

(b) What is the optimal face amount of life insurance, based on your answer from part (a)? Does your answer depend on your financial wealth or risk aversion?

(c) Is the optimal face amount of life insurance in part (b) more accurate for a younger or older person? Explain the reasoning for your answer.

(d) According to life-cycle theory, what share of your financial wealth should you allocate to stocks?

5. According to the term structure model, the t-year continuously compounded zero-coupon yield is

$$y(t) = \overline{y} + \frac{1}{t}\left(\frac{1-\rho^t}{1-\rho}\right)(y(1) - \overline{y}) + \frac{t-1}{t}\mu.$$

Assume that $\overline{y} = 5\%$, $\rho = 0.8$, and $\mu = 0.5\%$. The current 1-year continuously compounded zero-coupon yield is $y(1) = 5\%$. Assume no arbitrage.

[**20 points: 4 points for each part**]

(a) What are the 2- and 3-year continuously compounded zero-coupon yields?

(b) Which of the zero-coupon bonds from part (a) offers the higher expected return? Explain the reasoning for your answer.

(c) Is the term structure upward or downward sloping? Explain why the term structure is sloped that way.

(d) Consider a 3-year interest rate swap that exchanges a 1-year floating rate for a fixed rate. What is the fixed rate?

(e) Consider a futures contract on a 10-year zero coupon bond with face value of \$100. The futures contract matures in 1 year. What is the futures price?

6. Consider a stock that does not pay dividends, whose current price is \$150. The following tables reports prices of call options for various maturity and strike prices. For example, a call option on the stock with 2-month maturity and strike price of \$145 is trading at \$6.98. The continuously compounded riskless interest rate is 5% annually.

Maturity (months)	Strike (\$)	Call option price (\$)
2	145	6.98
2	150	4.10
4	145	
4	150	5.43

[**16 points: 4 points for each part**]

(a) What is the price of a put option on the stock with 4-month maturity and strike price of \$150?

(b) Based on the information given, what is the tightest lower bound on the price of the call option with 4-month maturity and strike price of \$145?

(c) Consider a protected put strategy where you simultaneously buy the stock and the put option from part (b). Draw the payoff diagram of the protected put strategy. A payoff diagram should have the overall payoff on the vertical axis and the price of the underlying stock on the horizontal axis. Clearly label the axes for full credit.

(d) Suppose that you buy a call option and a put option, both with the same maturity and a strike price of \$150. What is this strategy called? Write an equation that describes its payoff at maturity. What is this strategy betting on?